

Proposal for Choosing a Reasonable Number of LFR Communities for FADDIS

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Practical LFR rule for FADDIS [1]: choose the LFR community-size range so that the expected community size is approximately N/K for non-overlapping networks, or approximately $N[1 + f(O_m - 1)]/K$ for overlapping networks, where f is the fraction of overlapping nodes and O_m is the number of memberships per overlapping node.

- For $N = 100,000$, $K = 50, 100, 200$ corresponds to average community sizes of about 2,000, 1,000, and 500 nodes.
- For $N = 1,000,000$, $K = 50, 100, 200$ corresponds to average community sizes of about 20,000, 10,000, and 5,000 nodes.
- Use a community-size range such as:

$$c_{\min} = 0.4 \times \text{average size and } c_{\max} = 2 \times \text{average size.}$$

This keeps heterogeneity while avoiding thousands of communities.

Why is the classic LFR setting not well matched to FADDIS?

LFR is designed to generate benchmark graphs with heterogeneous degree and community-size distributions. Lancichinetti, Fortunato, and Radicchi introduced it because earlier benchmarks had unrealistically homogeneous node degrees and equal community sizes [2]. In the LFR construction, community sizes are drawn from a power-law-like distribution and the number of communities is usually an outcome of the selected size range, not a parameter chosen independently [2,3].

Classic LFR comparative experiments often used two small-to-medium community-size regimes: 10-50 nodes and 20-100 nodes, with community-size exponent $\tau_2 = 1$ [3].

Those settings are useful for comparability with the LFR literature, but they imply thousands or tens of thousands of communities when N is large. This is problematic for a sequential extractor such as FADDIS.

LFR formulas for controlling the community budget

Let N be the number of nodes, K the target number of planted communities, $c_{\min}$ and $c_{\max}$ the LFR community-size bounds, and τ_2 the community-size exponent.

In the LFR model, community sizes are drawn from a distribution proportional to $c^{-\tau_2}$ [2,3]. The expected community size is approximately:

$$\text{mean}(c) = [\sum c^{(1 - \tau_2)}] / [\sum c^{(-\tau_2)}], \text{ for } c = c_{\min}, \dots, c_{\max}$$

For a non-overlapping LFR network, the expected number of communities is approximately:

$$K \text{ approx. } N / \text{mean}(c)$$

For an overlapping LFR network, the total number of node-community memberships is larger than N . If O_n nodes overlap and each overlapping node has O_m memberships, then the total membership count is:

$$T = N + O_n(O_m - 1)$$

Therefore, for overlapping LFR networks:

$$K \text{ approx. } [N + O_n (O_m - 1)] / \text{mean}(c)$$

Equivalently, if $f = O_n/N$ is the fraction of overlapping nodes, choose:

$$\text{mean}(c) \text{ approx. } N [1 + f (O_m - 1)] / K$$

This membership-count logic follows from the overlapping extension of the LFR benchmark, where the total area covered by communities exceeds N when nodes can belong to multiple communities [3,4].

FADDIS oriented target values

Main recommendation. For a FADDIS study on large LFR networks, use a controlled community budget rather than the default small-community LFR regimes. The following target values are suggested:

Target K	Use	Interpretation
50	Main experiment	Very coarse macro-communities; useful for first large-scale test
100	Main experiment	Balanced default for sequential extraction
200	Main experiment	Upper main setting; still interpretable for FADDIS
500	Stress test	Use to evaluate scalability, not as the baseline
> 500	Not recommended as default	Becomes primarily a repeated-extraction throughput test

The exact number of communities returned by an LFR generator may vary slightly because community sizes are sampled. In reporting, state the target K , the generated K , the community-size bounds, and the random seed.

Concrete non-overlapping LFR settings

Use $c_{\min}$ approximately equal to 0.4 times the target average community size and $c_{\max}$ approximately equal to 2 times the target average community size.

This is a practical design heuristic: it preserves broad community sizes while keeping the number of planted communities within a FADDIS-compatible number

Target K	Mean size N/K	Suggested $c_{\min}$	Suggested $c_{\max}$	Role
50	2,000	800	4,000	Coarse baseline
100	1,000	400	2,000	Recommended default
200	500	200	1,000	Upper main setting
500	200	80	400	Stress test

Suggested non-overlapping LFR community-size bounds for $N = 1,000,000$.

Target K	Mean size N/K	Suggested $c_{\min}$	Suggested $c_{\max}$	Role
50	20,000	8,000	40,000	Coarse baseline
100	10,000	4,000	20,000	Recommended default
200	5,000	2,000	10,000	Upper main setting
500	2,000	800	4,000	Stress test

Check the LFR generator's feasibility conditions after choosing $c_{\min}$ and $c_{\max}$. In particular, $c_{\min}$ should be large enough to accommodate nodes' internal degree requirements. With the macro-community settings above, this is usually easy when average degree and maximum degree are in the common LFR ranges used in the literature [3].

Overlap settings

Use moderate overlap. For overlapping benchmarks, a defensible range is $f = O_n/N$ in $\{0.05, 0.10, 0.20, 0.30\}$ and O_m in $\{2, 3\}$. Xie, Kelley, and Szymanski report that a common feature observed by overlapping detection algorithms in real-world networks is a relatively small fraction of overlapping nodes, typically below 30%, with overlapping nodes usually belonging to only 2 or 3 communities [5]. Palla et al. also provide foundational empirical evidence that real networks often contain substantially overlapping communities [6].

Overlap fraction f	Memberships O_m	Multiplier $1 + f(O_m - 1)$	How to use it
0.10	2	1.10	Increase mean(c), c_min, and c_max by about 10% to keep K fixed
0.20	2	1.20	Increase mean(c), c_min, and c_max by about 20% to keep K fixed
0.30	2	1.30	Increase mean(c), c_min, and c_max by about 30% to keep K fixed
0.30	3	1.60	Increase mean(c), c_min, and c_max by about 60% to keep K fixed

Example. For $N = 100,000$, $K = 100$, $f = 0.20$, and $O_m = 2$, the target average community size becomes $100,000 \times 1.20 / 100 = 1,200$. A compatible range would be c_min about 480 and c_max about 2,400.

Observations in the context of FADDIS

1. Because FADDIS extracts clusters sequentially, the standard LFR regimes with many small communities can imply thousands of planted communities for large N , making the benchmark dominated by repeated extraction cost.

Therefore, we use LFR benchmarks with a controlled planted community budget K in $\{50, 100, 200\}$, and $K = 500$ as a scalability stress test. The LFR community-size bounds were selected so that the expected community size was approximately N/K for non-overlapping networks and $N[1 + f(O_m - 1)]/K$ for overlapping networks. This retains the LFR principle of heterogeneous community sizes while aligning the benchmark with the sequential nature of FADDIS [1-4].

2. Using $K = 50$ -200 on $N = 100,000$ or $N = 1,000,000$ creates a macro-community benchmark. This is not the same regime as the classic LFR small-community benchmark. It is appropriate for testing FADDIS under large- N conditions, but it should be described transparently as a large-community LFR scenario. Leskovec, Lang, Dasgupta, and Mahoney found that tight communities in many large social and information networks often occur at small size scales, while larger communities can blend into the network core and become less sharply community-like [7]. Thus, increasing community sizes is computationally sensible for FADDIS, but it changes the scientific regime being tested.

References

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